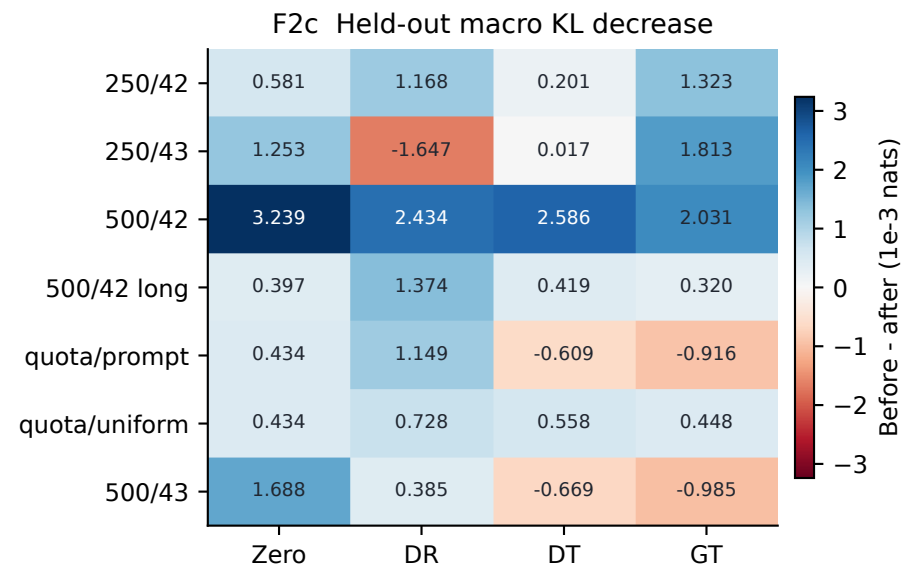
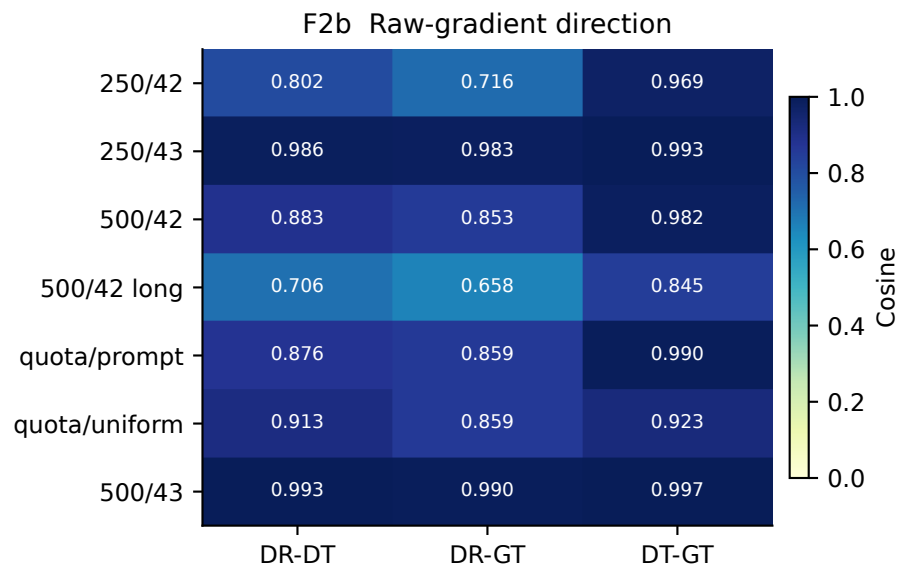
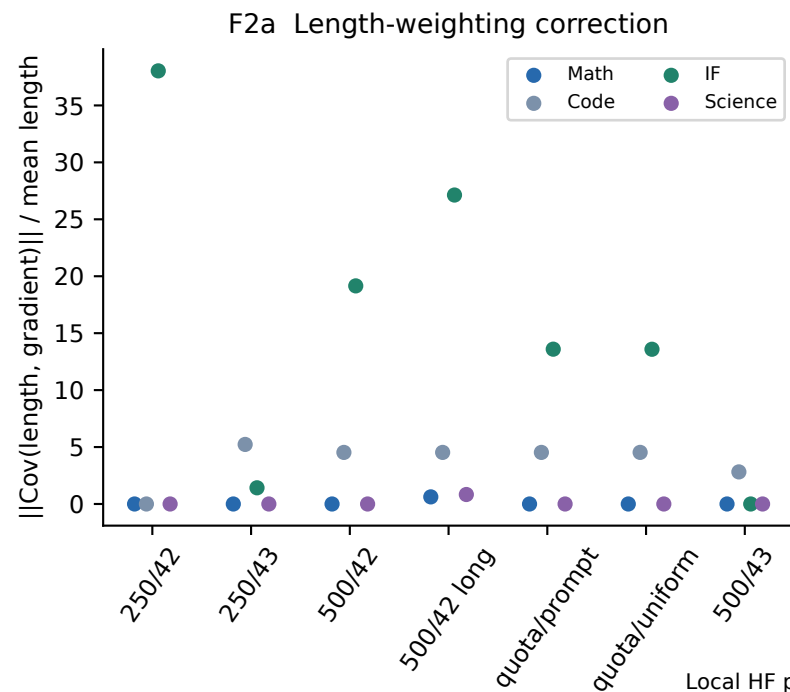


F2 | Normalization changes local gradients

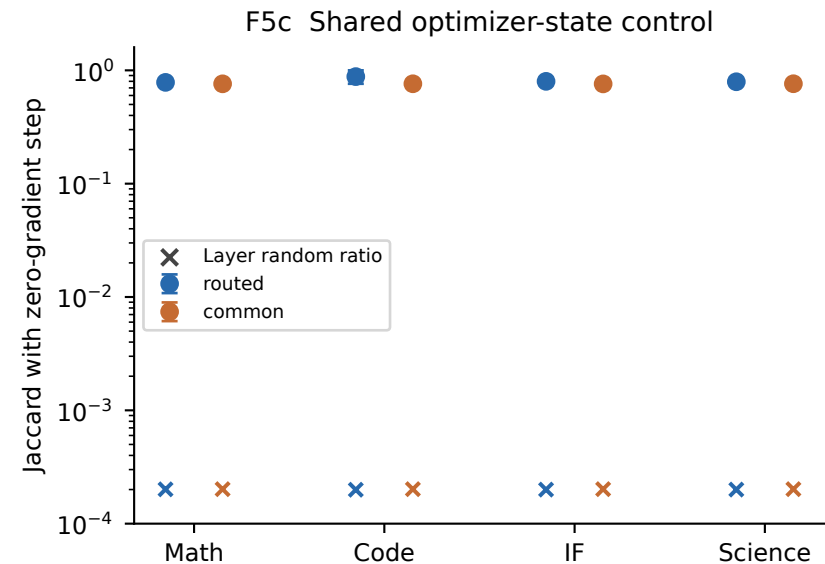
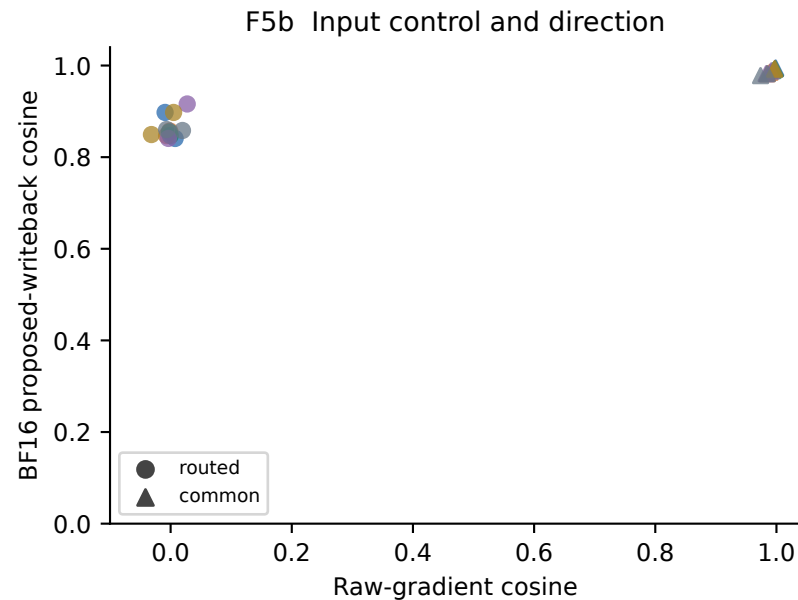
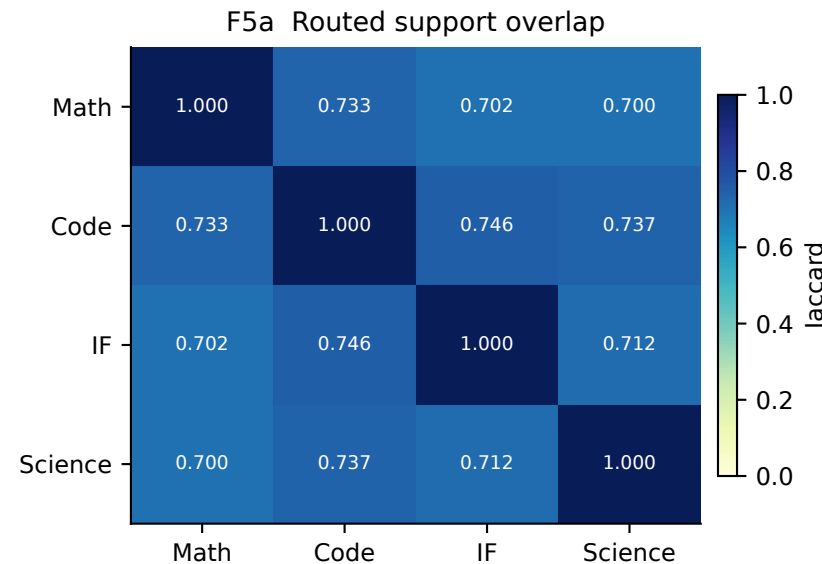
All seven Overlap64-loss probes. KL panel is exploratory: one held-out response/domain; small changes and repeat-run numerical differences.



Local HF probes with saved Adam state; BF16 proposed writebacks, not online steps. Diagnostic banks are not training seeds.

F5 | Teacher updates at M-PG/250 | PG

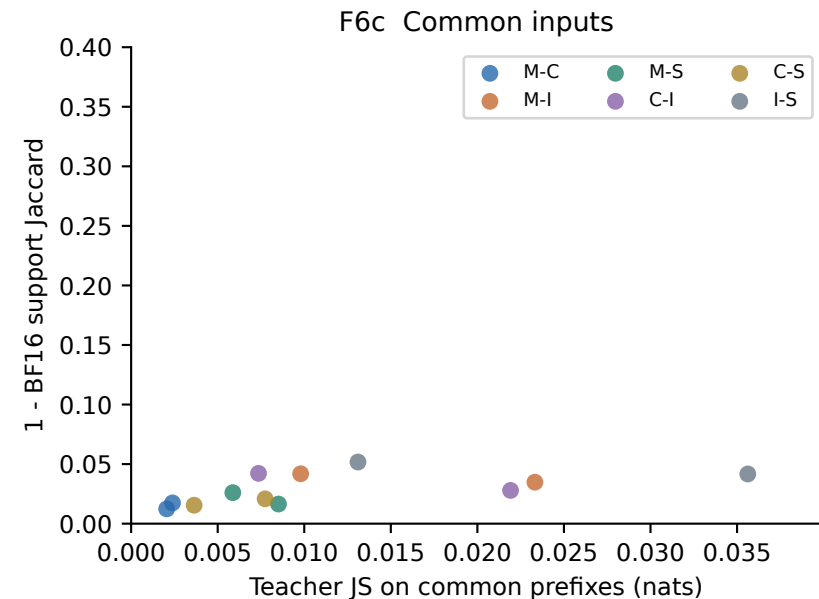
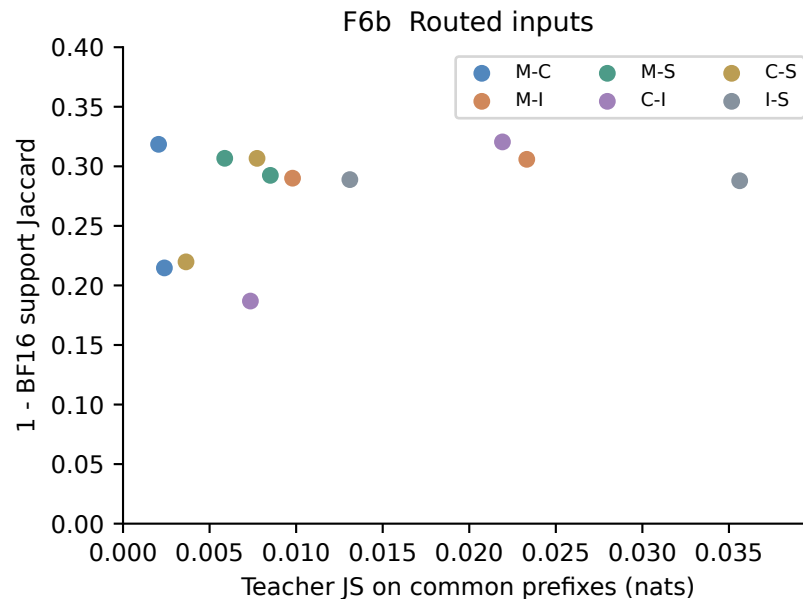
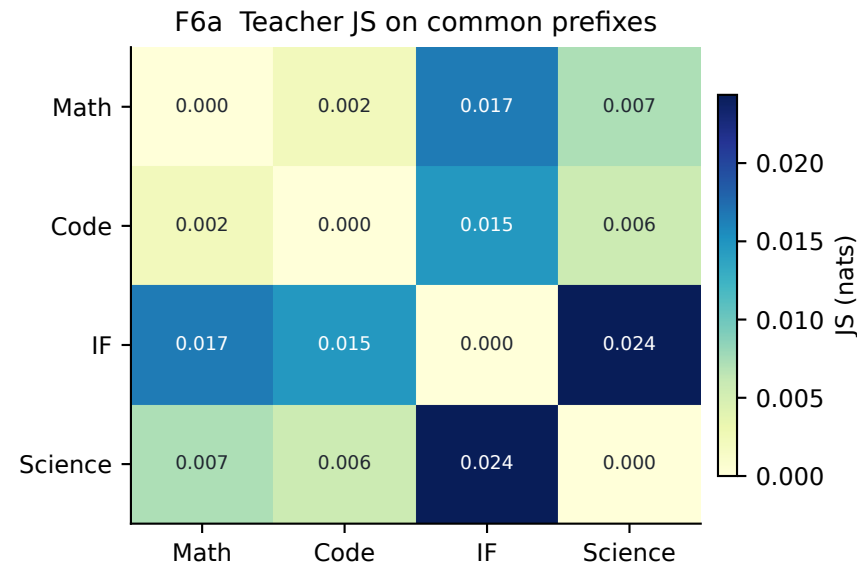
Two diagnostic banks (42/43); six shared teacher pairs/bank. Support threshold = $1e-5$. Error bars show observed bank min-max, not confidence intervals.



Local HF probes with saved Adam state; BF16 proposed writebacks, not online steps. Diagnostic banks are not training seeds.

F6 | Teacher distribution distance at M-PG/250 | PG

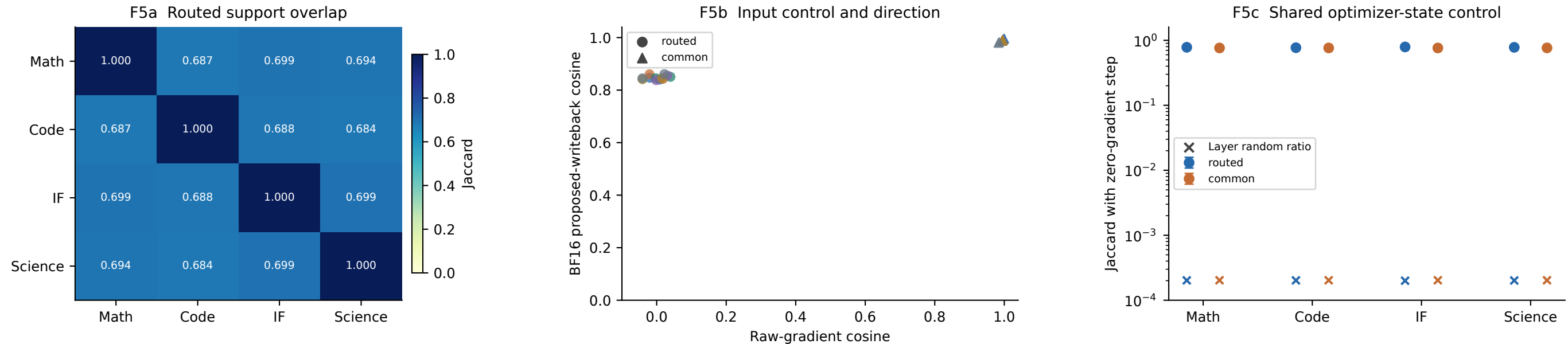
JS uses common four-domain prefixes. Each point is a bank/pair measurement; teacher pairs are dependent. No regression or significance claim.



Local HF probes with saved Adam state; BF16 proposed writebacks, not online steps. Diagnostic banks are not training seeds.

F5 | Teacher updates at M-PG/250 | Overlap64

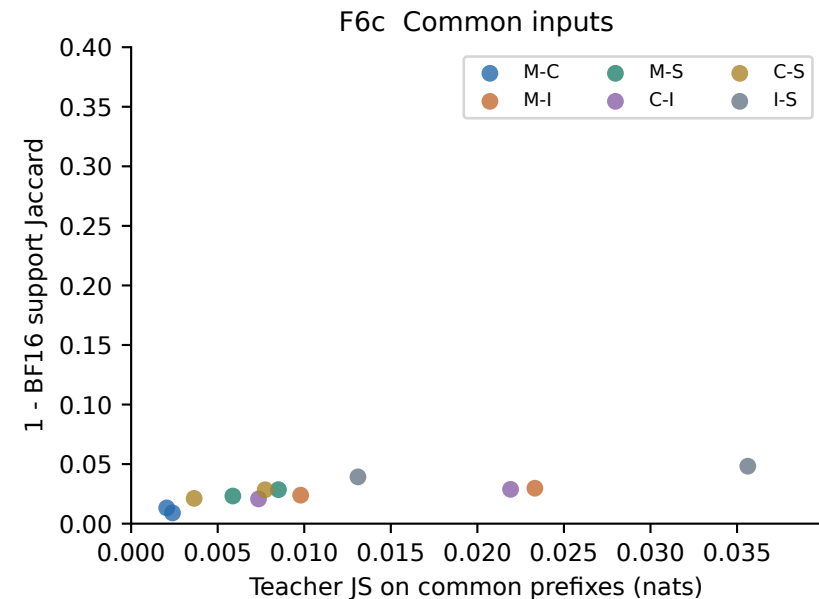
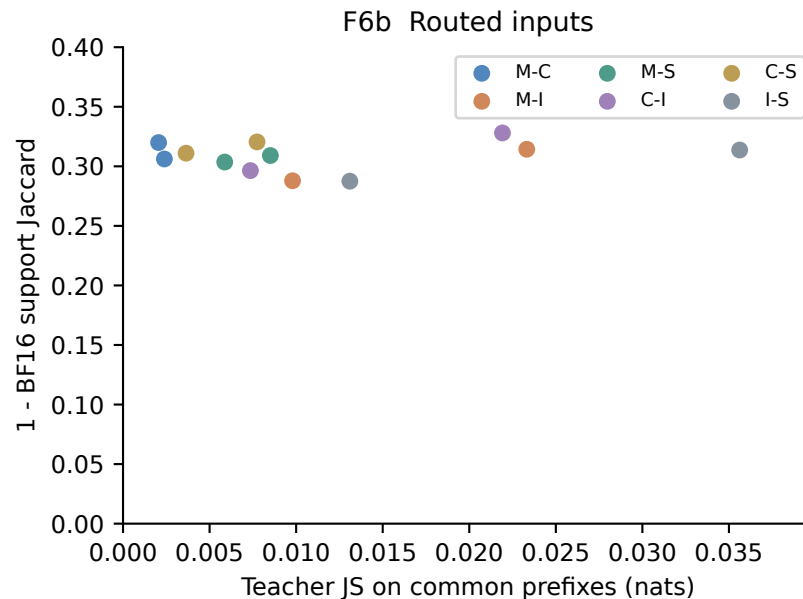
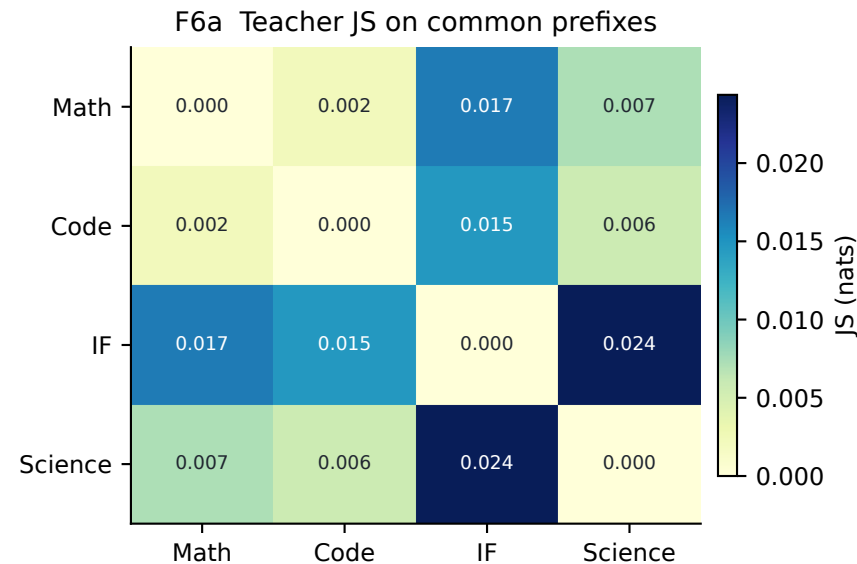
Two diagnostic banks (42/43); six shared teacher pairs/bank. Support threshold = $1e-5$. Error bars show observed bank min-max, not confidence intervals.



Local HF probes with saved Adam state; BF16 proposed writebacks, not online steps. Diagnostic banks are not training seeds.

F6 | Teacher distribution distance at M-PG/250 | Overlap64

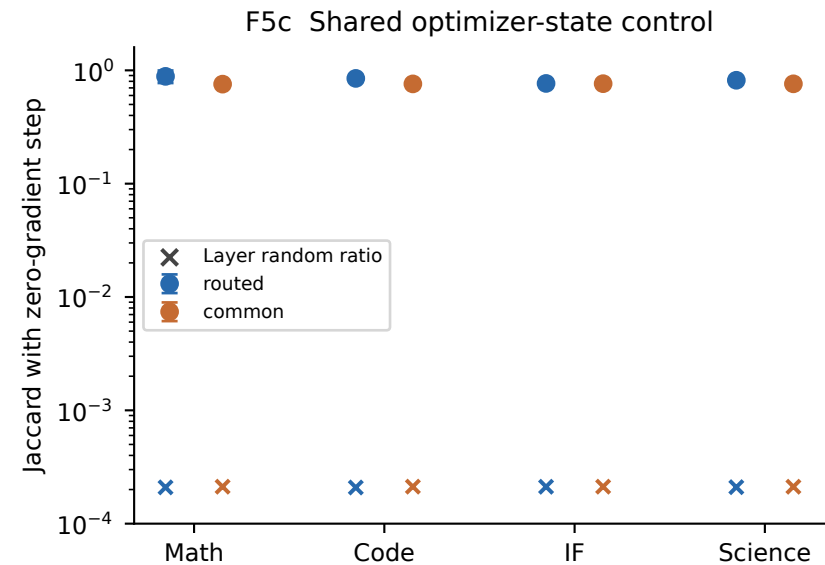
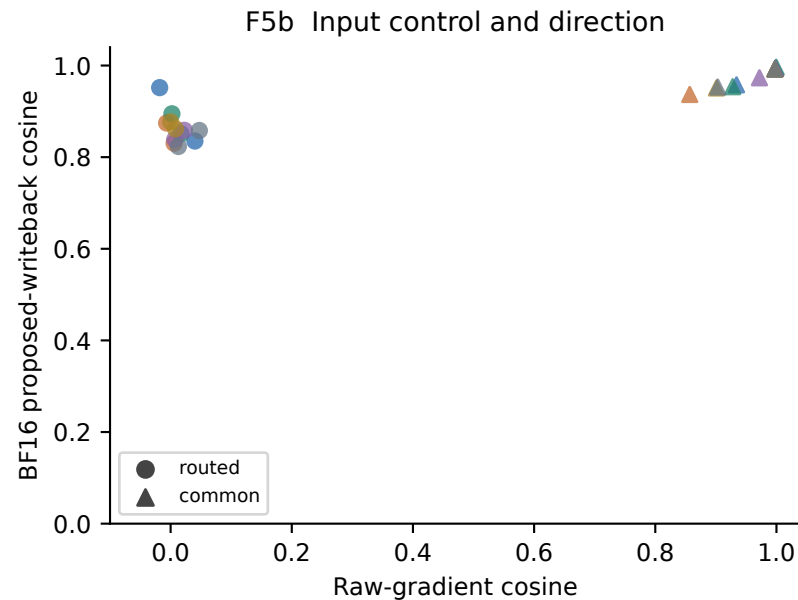
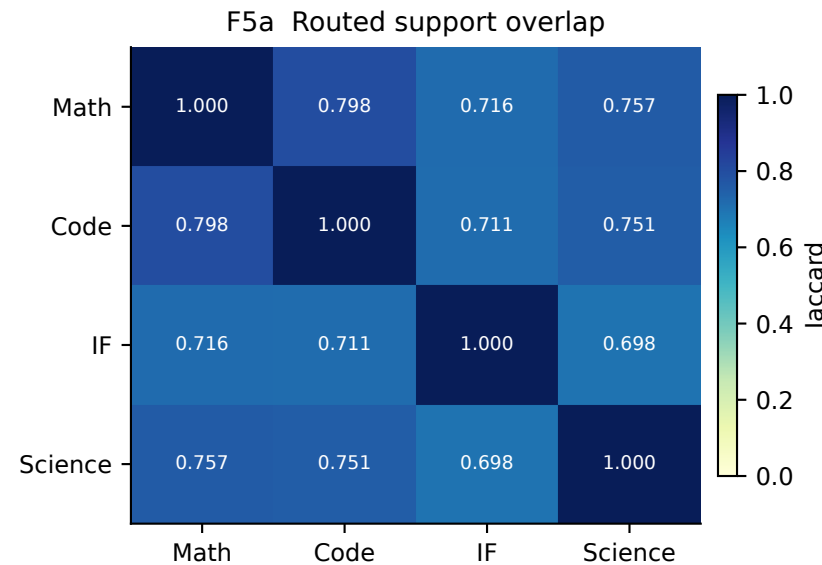
JS uses common four-domain prefixes. Each point is a bank/pair measurement; teacher pairs are dependent. No regression or significance claim.



Local HF probes with saved Adam state; BF16 proposed writebacks, not online steps. Diagnostic banks are not training seeds.

F5 | Teacher updates at M-PG/500 | PG

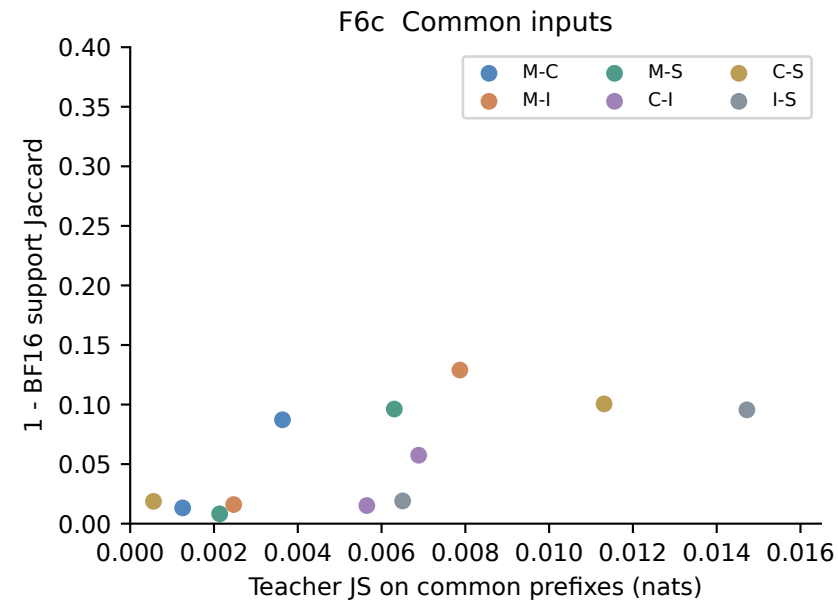
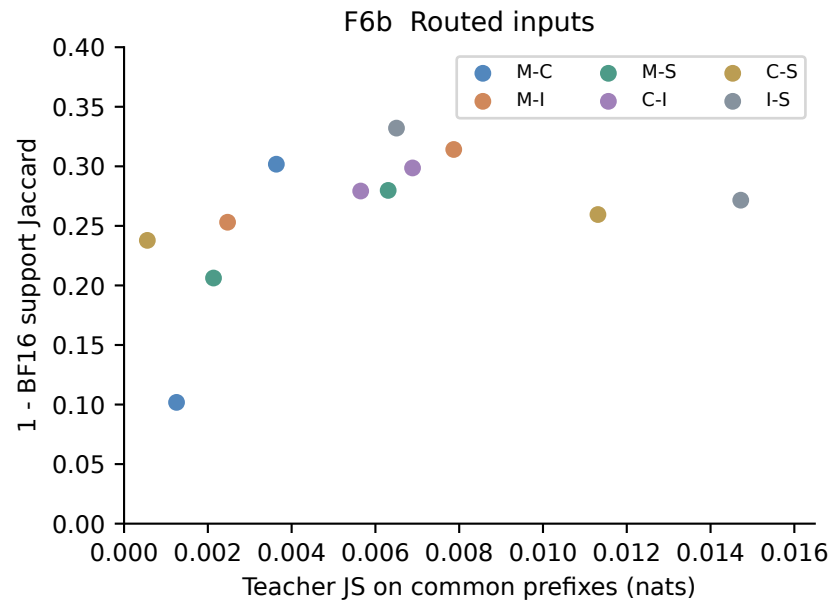
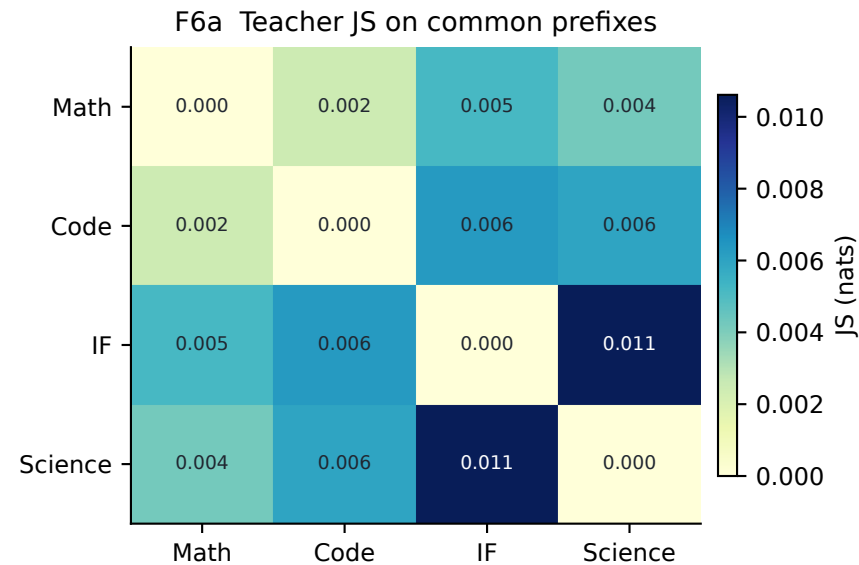
Two diagnostic banks (42/43); six shared teacher pairs/bank. Support threshold = $1e-5$. Error bars show observed bank min-max, not confidence intervals.



Local HF probes with saved Adam state; BF16 proposed writebacks, not online steps. Diagnostic banks are not training seeds.

F6 | Teacher distribution distance at M-PG/500 | PG

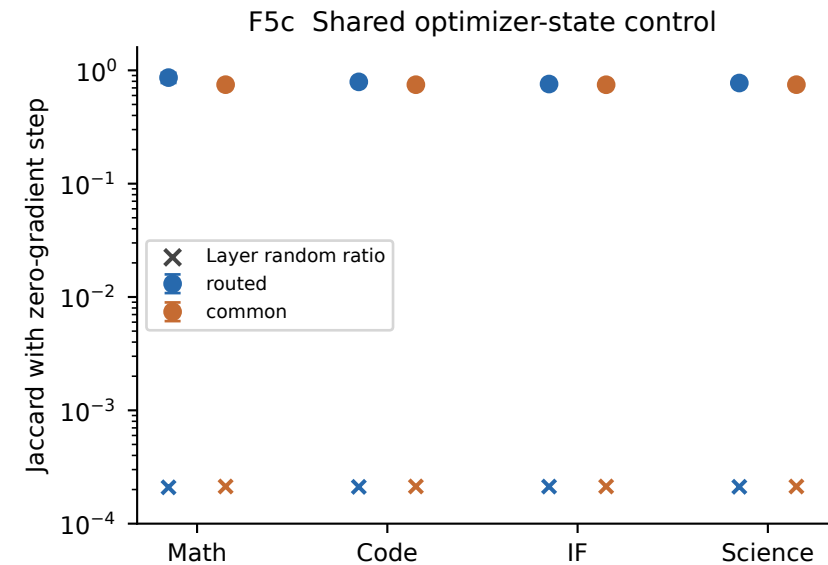
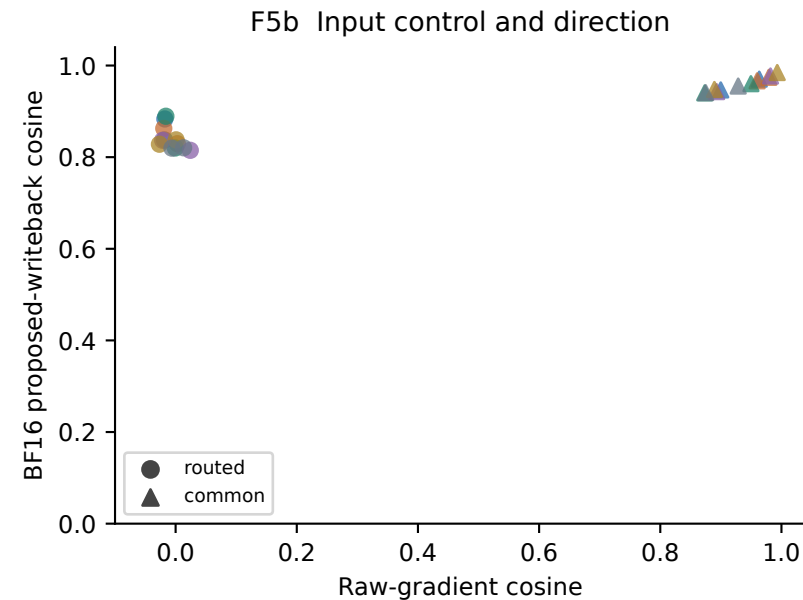
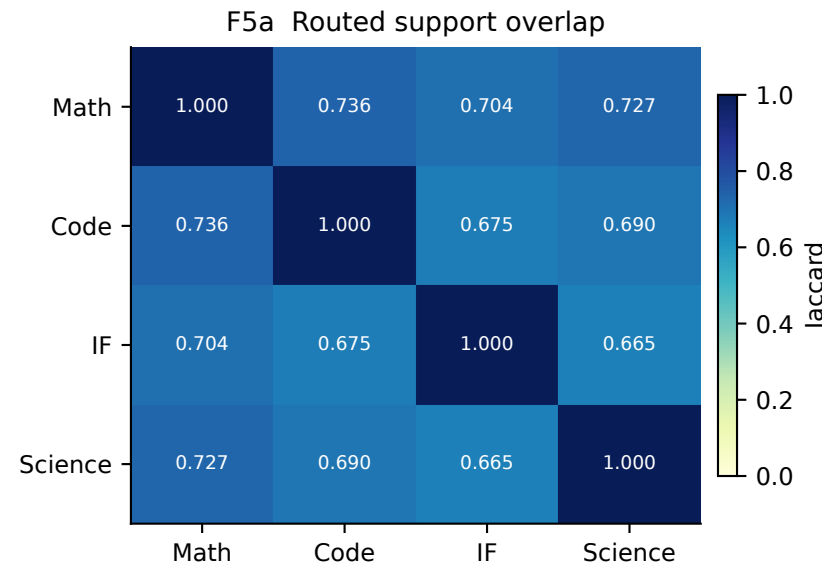
JS uses common four-domain prefixes. Each point is a bank/pair measurement; teacher pairs are dependent. No regression or significance claim.



Local HF probes with saved Adam state; BF16 proposed writebacks, not online steps. Diagnostic banks are not training seeds.

F5 | Teacher updates at M-PG/500 | Overlap64

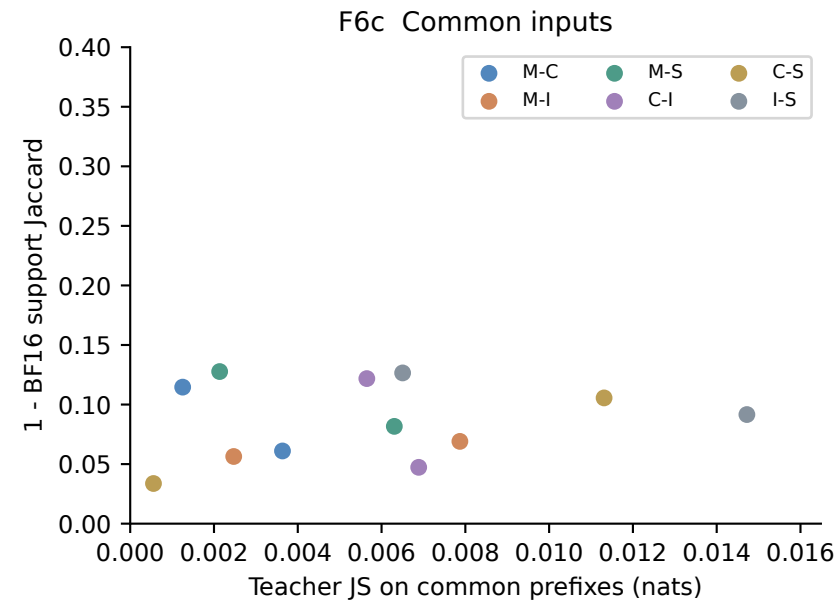
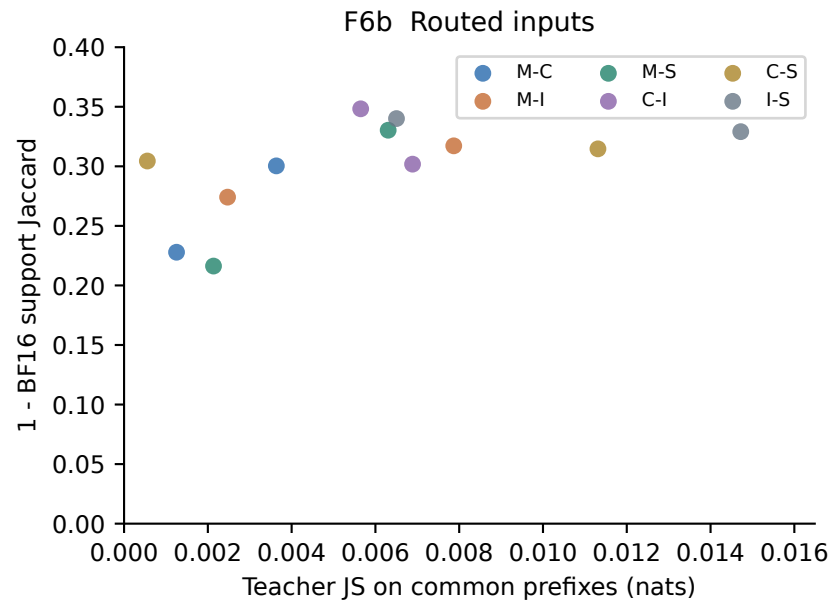
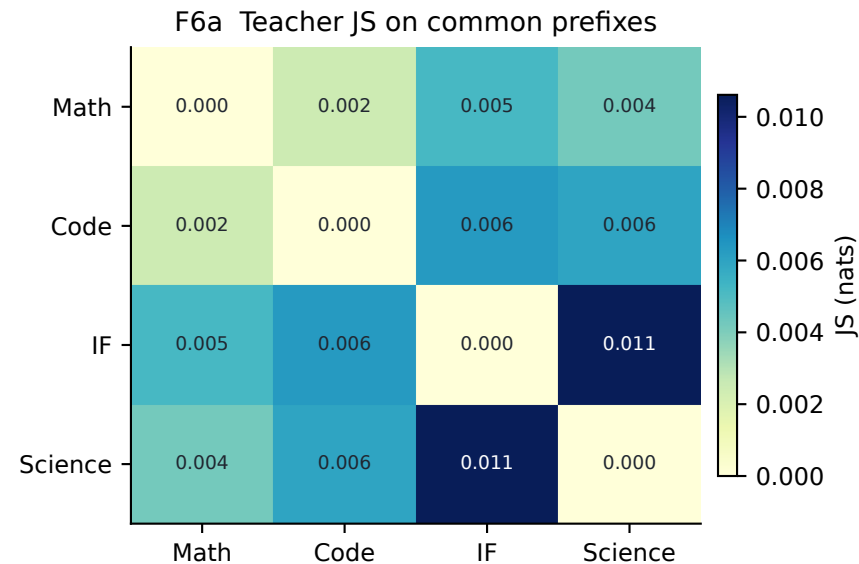
Two diagnostic banks (42/43); six shared teacher pairs/bank. Support threshold = $1e-5$. Error bars show observed bank min-max, not confidence intervals.



Local HF probes with saved Adam state; BF16 proposed writebacks, not online steps. Diagnostic banks are not training seeds.

F6 | Teacher distribution distance at M-PG/500 | Overlap64

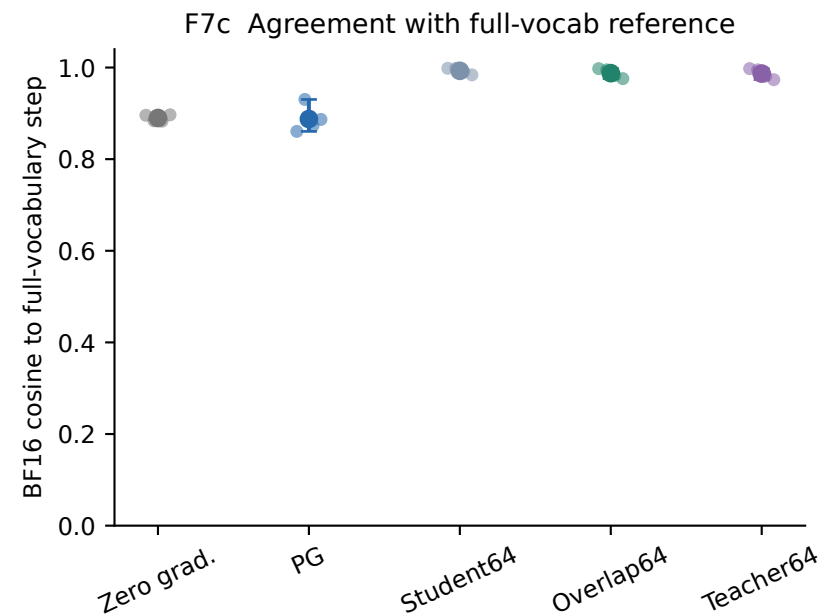
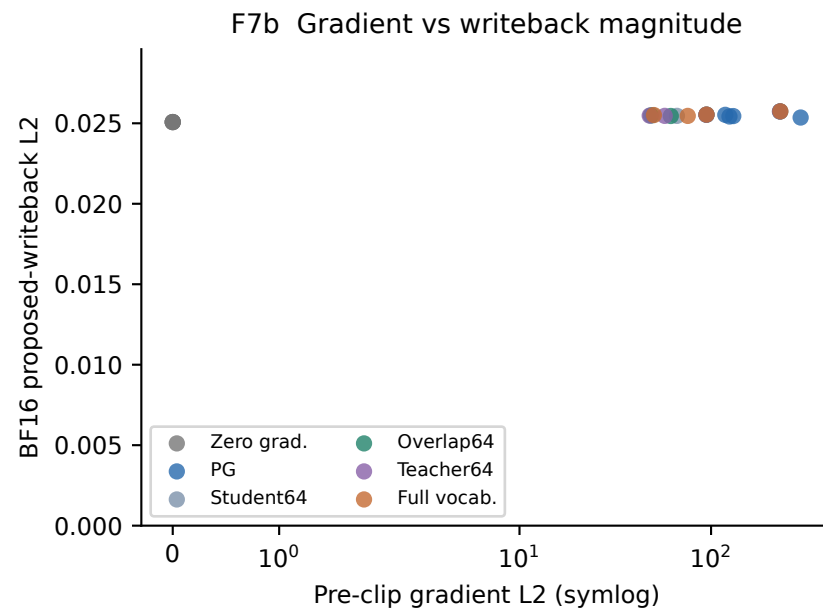
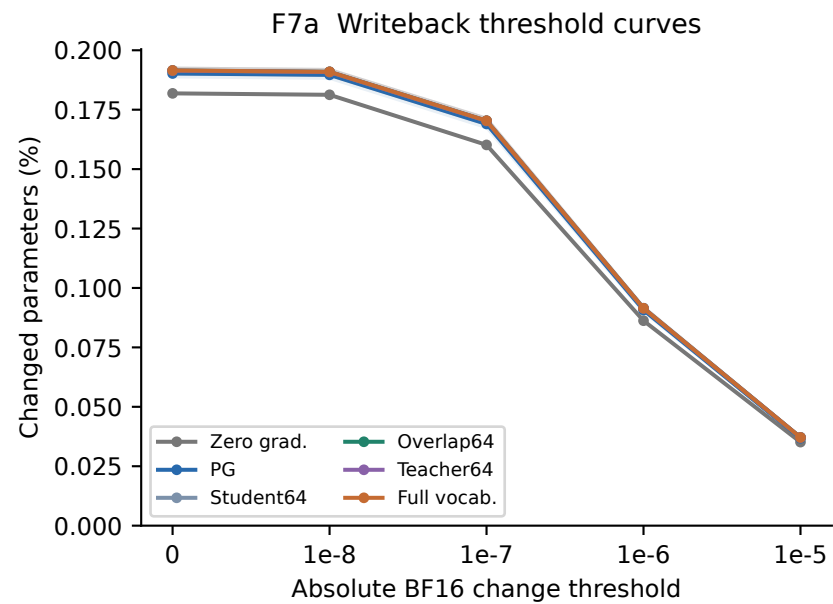
JS uses common four-domain prefixes. Each point is a bank/pair measurement; teacher pairs are dependent. No regression or significance claim.



Local HF probes with saved Adam state; BF16 proposed writebacks, not online steps. Diagnostic banks are not training seeds.

F7 | Local supervision controls | m-pg250

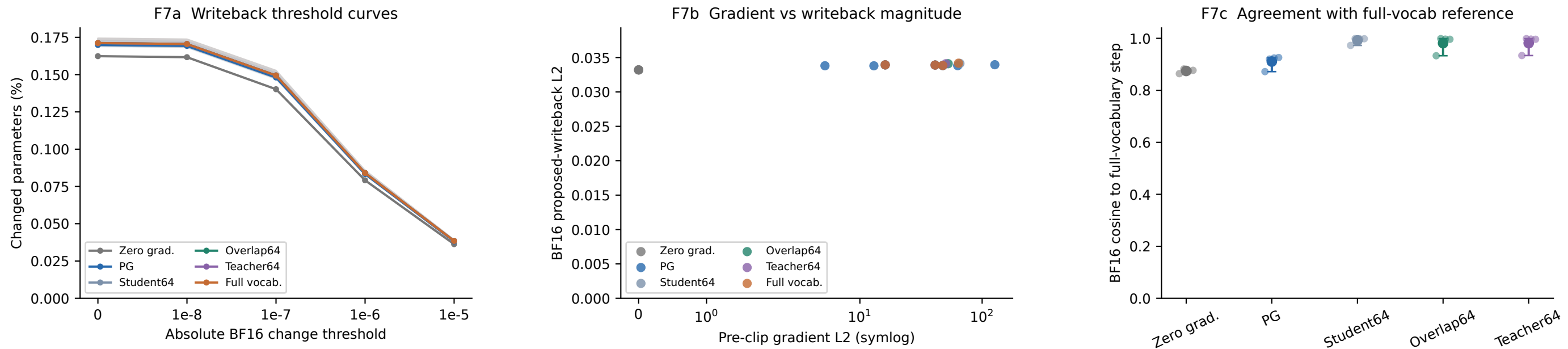
Four banks (42-45); cap256, up to four prefixes/response. Bands/bars are bank min-max. Different losses change weighting as well as support.



Local HF probes with saved Adam state; BF16 proposed writebacks, not online steps. Diagnostic banks are not training seeds.

F7 | Local supervision controls | m-pg500

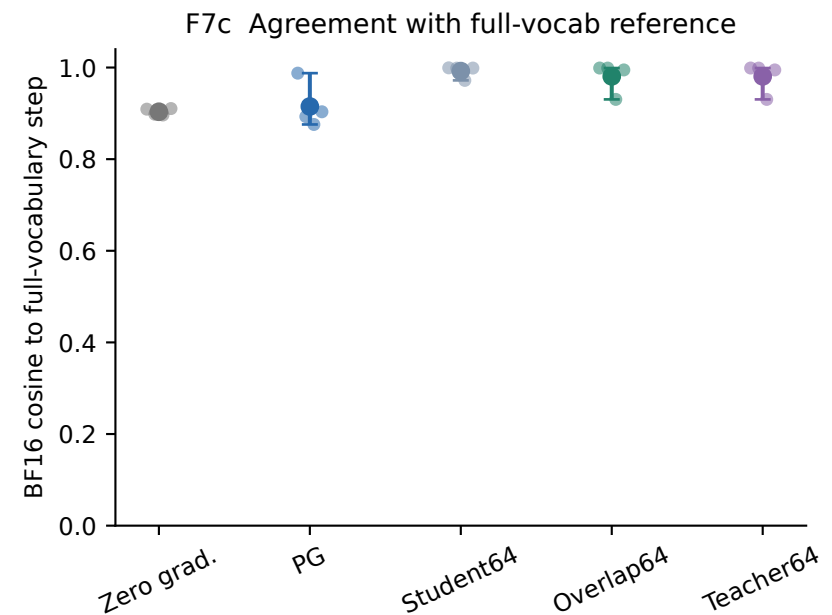
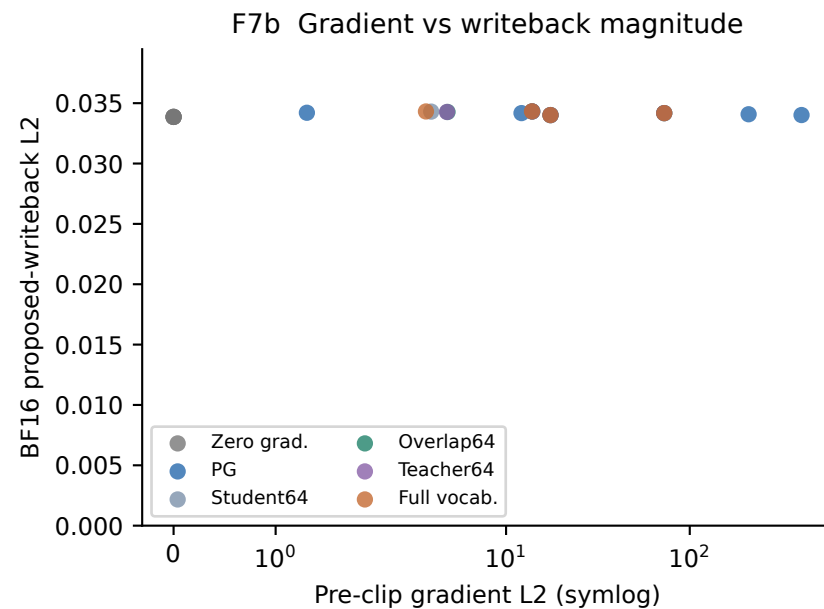
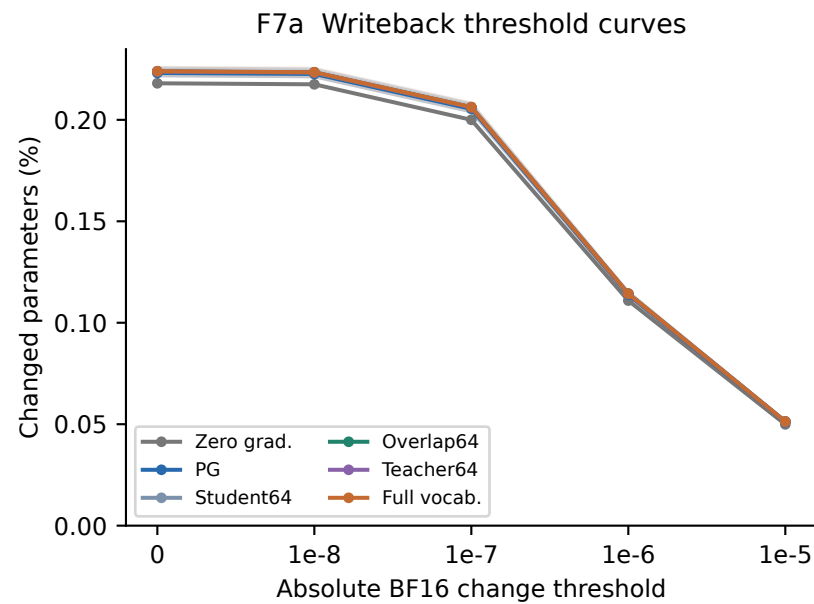
Four banks (42-45); cap256, up to four prefixes/response. Bands/bars are bank min-max. Different losses change weighting as well as support.



Local HF probes with saved Adam state; BF16 proposed writebacks, not online steps. Diagnostic banks are not training seeds.

F7 | Local supervision controls | s-pg250

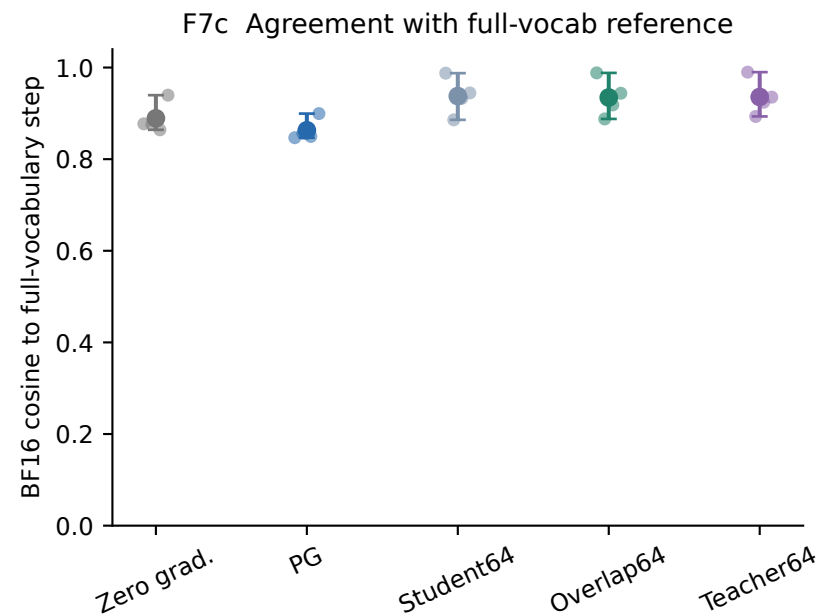
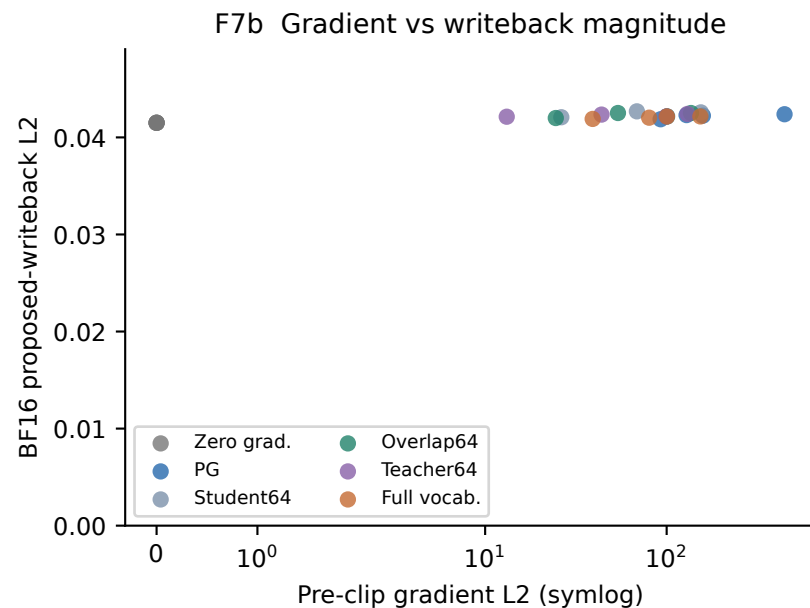
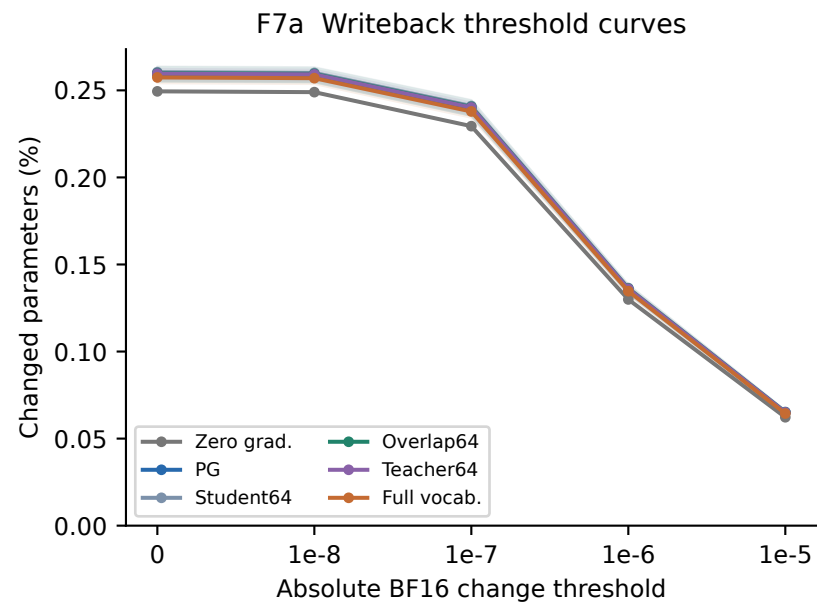
Four banks (42-45); cap256, up to four prefixes/response. Bands/bars are bank min-max. Different losses change weighting as well as support.



Local HF probes with saved Adam state; BF16 proposed writebacks, not online steps. Diagnostic banks are not training seeds.

F7 | Local supervision controls | m-i64dr250 | mixed Student64 -> Overlap64 training history

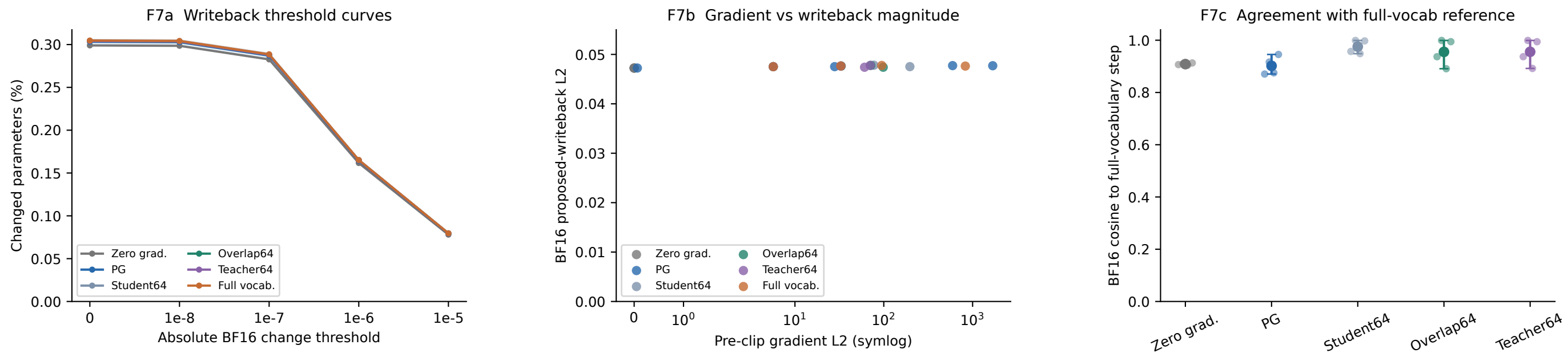
Four banks (42-45); cap256, up to four prefixes/response. Bands/bars are bank min-max. Different losses change weighting as well as support.



Local HF probes with saved Adam state; BF16 proposed writebacks, not online steps. Diagnostic banks are not training seeds.

F7 | Local supervision controls | s-i64250 | mixed Student64 -> Overlap64 training history

Four banks (42-45); cap256, up to four prefixes/response. Bands/bars are bank min-max. Different losses change weighting as well as support.



Local HF probes with saved Adam state; BF16 proposed writebacks, not online steps. Diagnostic banks are not training seeds.